INCH-POUND

MIL-PRF-1/1575B
26 April 2005
SUPERSEDING
MIL-E-1/1575A(NAVY)
10 August 1970

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 10VP7

This specification sheet is inactive for new design after 8 May 2000.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Magnetic deflection and focus.

Dimensions and pin connections: See figure 1.

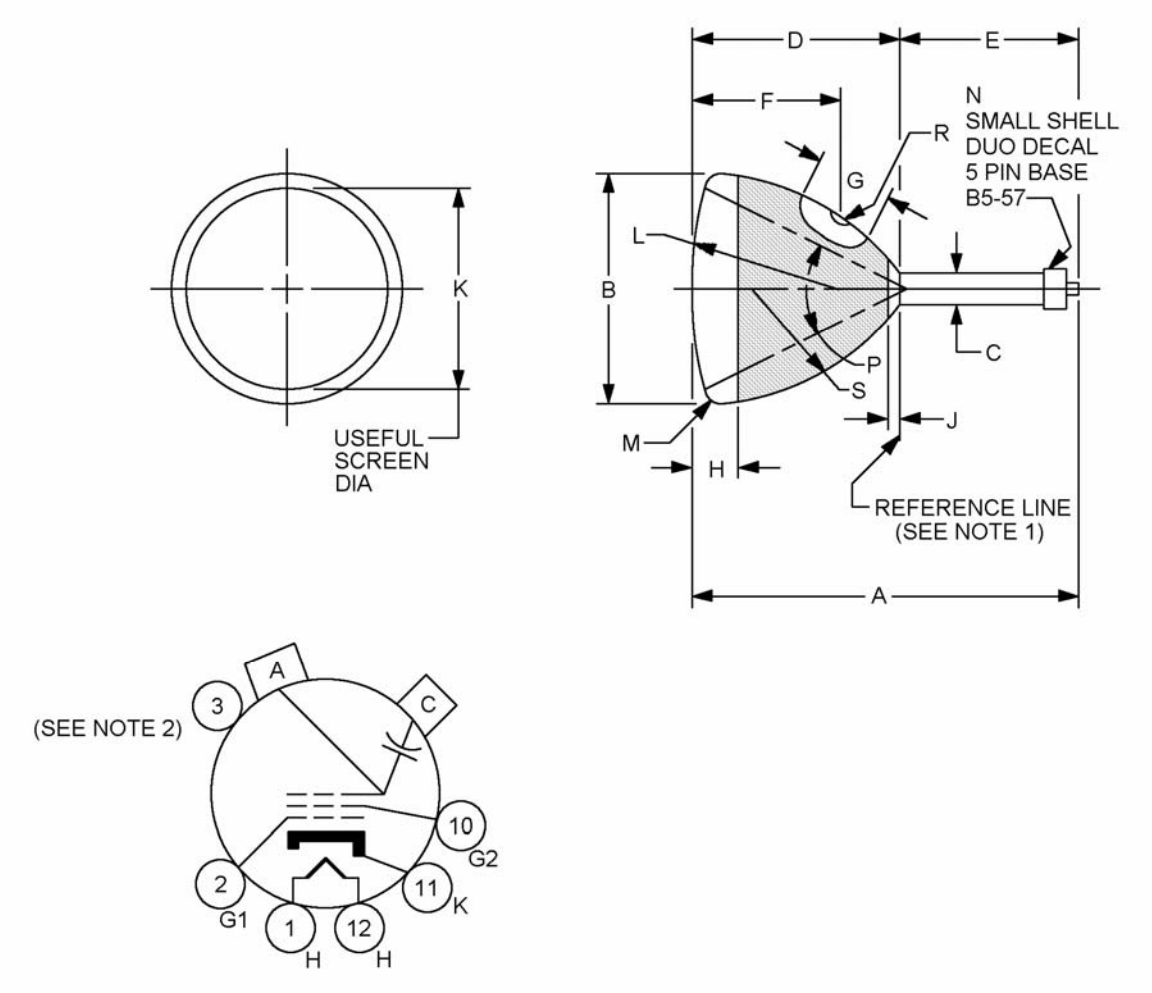
ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt.
Unit:	V	V dc	V dc	V dc	V dc	Meg Ω	ft.
Maximum:	6.9	0-140 (see note 1)	450	13,200	±154	1.5	60,000
Minimum:	5.7	---	---	11,000	---	---	---
Test conditions:	6.3	Adjust	250	11,000	---	---	---

GENERAL:

Qualification: Not required.

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Pin connections
1. Heater
2. Grid 1
10. Grid 2
11. Cathode
12. Heater
Cap anode

NOTE 1: Reference line is determined by position where JETEC No. 112 reference line gauge will seat against bulb.
NOTE 2: Vacant position pin No. 3 aligned with anode contact cap (J1-21) within 30 degrees.

FIGURE 1. Outline drawing for tube type 10VP7.

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Ltr	Qualification	
N R	Base: B5-57 Cap: J1-21	
Conformance inspection, part 2		
Ltr	Min	Max
A	17.25	18.00
B	10.38	10.63
C	1.38	1.50
D	9.25	9.63
E	8.00	8.38
F	6.87	7.63
G	4.0	4.5
H	2.25	2.75
J	.5	1.0
K	9.13	---
Ltr	Nominal dimensions	
L	42R	
M	7/16R	
P	50°	
S	20R	

FIGURE 1. Outline drawing for tube type 10VP7 - Continued.

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TABLE I. Test conditions.

MIL-STD-1311	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
Method	Qualification					
1141	Pressure	---	---	---	---	---
1002	Barometric pressure, reduced	Pressure = 54 mmHg	---	---	---	---
5111	Vibration (cathode-ray tubes)	---	Width	---	2.0	mm
1331	Direct interelectrode capacitance	Cathode to all g to all	C C	--- ---	5.0 6.0	pF pF
5201	Electrode currents Grid No. 2	Ec1 = 0	Ic2	---	25.0	µA dc
5216	Cathode illumination	----	---	---	---	---
1216	Base material insulating quality	---	---	---	---	---
	Conformance inspection, part 1					
5201	Voltage breakdown	---	---	---	---	---
5201	Voltage breakdown (magnetic types)	Ec1 Adjust for cutoff -28 to -72 V dc Ec2 = 250 V dc Eb = 11,000 V dc	---	---	---	---
5206	Gas ratio	---	Gr	---	0.25	---
5106	Blemishes	---	---	---	---	---
5231	Spot position	---	---	---	18	mm
5236	Zero-bias anode current	---	---	---	---	---
5241	Grid-cutoff voltage	---	ECL	-33	-77	V dc

See footnotes at end of table.

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TABLE I. Test conditions - Continued.

MIL-STD-1311	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
Method	Conformation inspection part 1 - Continued.					
5246	Focusing ampere turns	---	AT	100	120	---
5251	Grid No. 1 leakage current	---	---	---	---	---
---	Resolution, center	Ib = 200 µA dc; (see note 4)	Res.	300	---	lines
5101	Neck straightness	(See note 2)	---	---	---	---
---	Aperture alignment	(See note 5)	Distance	---	8.0	mm
5223	Modulation	Ib = 200 µA dc	ΔEc1	---	55	V dc
	Conformance inspection, part 2					
1101	Base, cap, and insert secureness	---	---	---	---	---
1301	Heater current	---	If	530	660	mA
5101	Side terminal and base (alignment)	Pin position No. 3	---	---	---	---
5101	Neck and bulb alignment	---	---	---	---	---
5101	Face tilt	---	---	---	---	---
5216	Stray emission	Eb = 11,000 V dc Ec2 = 450 V dc	---	---	---	---
5221	Screen	(See note 3)	---	---	---	---
5251	Heater-cathode leakage	---	---	---	---	---
5251	Grid No. 2 leakage current	---	---	---	---	---

See footnotes at end of table.

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TABLE I. Test conditions - Continued.

MIL-STD-1311	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
Method	Conformation inspection part 3					
---	Life test	Group C; Eb = 11,000 V dc; Ec2 = 450 V dc; Ib = 60 µA dc	t	1,000	---	hrs
---	Life test end points	Resolution, center modulation Grid No. 1 leakage current Heater-cathode leakage Grid No. 2 leakage current Stray emission	Res. ΔEc1	200 ---	--- 55	lines V dc

NOTES:

1. When Ec2 is greater than 330 V dc, or Ec1 is near zero, or both, the effective resistance of the anode and grid No. 3 supply shall be adequate to limit the anode and grid No. 3 input power to 6 watts. The peak grid No. 1 drive from cutoff shall not exceed 65 volts.

2. Neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5 inches long and 1.5000 +.003, -.000 inches inside diameter. This cylinder shall move freely between the reference line and the base of the assembled tube.

3. The test conditions shall be anode voltage (relative to cathode) 4,000 V minimum and beam current 60 µA maximum.

4. Use raster scanning pattern 6 inch by 8 inch composed of 525 lines interlaced and synchronized. With Ec1 at cutoff, vary signal from video generator utilizing a 2F21 monoscope from zero to indicated anode current value. Focus for equal horizontal and vertical resolution in the center of the pattern.

5. The distance between the center of the unfocused, un-deflected spot at a low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, un-deflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 10 seconds to prevent damage to the screen.

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NOTE: Referenced documents. In addition to MIL-PRF-1, this document references the following:

MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - EC
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH

Preparing activity:
DLA - CC

(Project 5960-3758)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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